The Hidden Cost of Switching

Working paper on attention, interruption and reading comprehension

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1. Introduction

Readers rarely move through a difficult text in one sitting. They pause to look up a term, check a message, or open a second source, and each pause seems small. This paper asks what those pauses cost, measured not in minutes but in how much of the text the reader can later recall and use.

Earlier work on task switching measured the delay in starting a new task. We focus on the return trip: what happens to comprehension when a reader comes back to a paragraph they left half-understood.

Our central claim is that the cost of an interruption depends less on its length than on whether the reader leaves the page. A thirty-second definition shown beside the text costs little. The same definition found in another tab costs far more, because the reader must rebuild their place in the argument.

2. Method

We recruited 120 graduate students and asked each to read two passages of about 2,500 words from introductory economics and distributed systems texts. Each passage contained six terms that most participants would not know.

In the in-place condition, a definition appeared in a side panel when the reader selected a term. In the switch condition, readers searched for the term in a separate browser tab. Reading time, the number of re-read sentences, and a recall test 48 hours later were recorded.

Re-reading was measured from scroll position: a sentence counted as re-read when it returned to the middle third of the screen after having left it for more than five seconds.

3. Results

Readers in the switch condition re-read 2.4 times as many sentences after each lookup. Total reading time rose by 19 percent, most of it spent finding their place rather than reading new material.

Recall after 48 hours was 14 percent lower in the switch condition. The effect was largest for questions that required connecting two paragraphs, and smallest for questions about isolated facts.

Participants underestimated the cost. Asked afterwards, most said the lookups had taken "a few seconds" and had not affected their understanding.

4. Discussion

The results suggest a design principle for reading tools: help should arrive where the reader already is. Any feature that sends the reader elsewhere pays a re-entry cost that readers do not notice and do not report.

A second implication concerns notes. Readers who wrote a one-line note at the moment of confusion recovered their place faster, even when they did not return to the note later. The act of writing appears to mark the spot in memory as well as on the page.

Limitations: the passages were short, and the 48-hour test cannot tell us whether the effect lasts. A longer study with real reading lists is under way.